

Part B-1

1. Excellence

1.1 Quality and pertinence of the project's research and innovation objectives (and the extent to which they are ambitious, and go beyond the state of the art)

Critical infrastructures, including water, energy, industrial, transport and digital systems, operate as interconnected cyber-physical networks monitored through heterogeneous sensors, actuators, control platforms and communication devices. Abnormal behaviour may result from equipment degradation, sensor malfunction, communication disruption, unexpected operating conditions or deliberate manipulation. Existing anomaly detectors can flag deviations, but they rarely determine where an anomaly originates, how it relates to connected assets, how it may propagate, how reliable the evidence is, or whether it is more consistent with a fault, communication disturbance or potentially malicious event. Anomaly scores should therefore support investigation and prioritisation, not be treated as conclusive cyberattack attribution.

Functional Data Analysis (FDA) and functional statistical process monitoring provide interpretable representations and statistically calibrated monitoring of densely sampled trajectories [1,2]. Deep anomaly-detection methods capture nonlinear temporal dependencies but remain challenged by scarce labels, changing operating regimes and poorly calibrated alarms [3]. Graph-based approaches explicitly model inter-variable dependencies and support graph-aware anomaly detection and localisation in multivariate time series [4,5]. Recent topology- and physics-aware graph methods for water cyber-physical systems further combine localisation, robustness analysis and explainability [7], while BATADAL provides an established benchmark for attack detection [6]. Explainability methods can identify influential variables or graph structures, but plausible explanations are not necessarily reliable or stable [8]. However, these approaches do not yet integrate FDA-based continuous-trajectory representation, calibrated uncertainty, reliability-gated explanations, propagation evidence and human-supervised investigation prioritisation within one reproducible framework. This specific integration gap motivates Trustworthy Graph-Functional AI for Cyber-Physical Anomaly Detection in Critical Infrastructures (GRAIL-ORI).

GRAIL-ORI addresses this gap by developing a trustworthy graph-functional anomaly-intelligence framework. Functional representations and statistically calibrated monitoring evidence will be embedded in infrastructure graphs; self- and semi-supervised models will detect and localise anomalous nodes, edges and subgraphs; uncertainty and explanation reliability will be quantified; and an Operational Risk Intelligence (ORI) layer will convert validated evidence into affected-component information, propagation paths and ranked investigation priorities under human oversight. The central research question is: How can continuous operational trajectories and graph-structured dependencies be jointly modelled to provide accurate, calibrated, explainable and operationally useful anomaly intelligence under scarce labels, noisy measurements and evolving infrastructure conditions? The overall objective is to develop and validate this framework across two complementary cyber-physical water-system benchmarks while establishing its robustness, limitations and conditions for transfer.

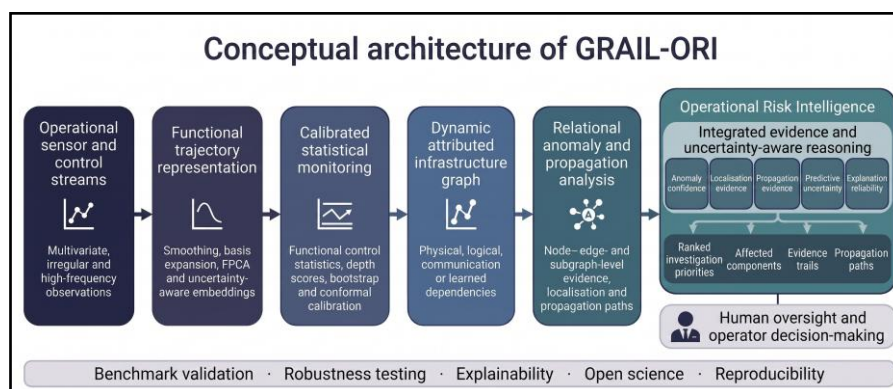


Figure 1. Conceptual architecture of GRAIL-ORI. The framework transforms functional operational trajectories and infrastructure dependencies into calibrated, explainable and uncertainty-aware anomaly evidence, which is converted into human-supervised Operational Risk Intelligence.

Objective	Research content	Measurable and verifiable outputs	Implementation
O1: Calibrated functional monitoring	Develop functional representations and statistically calibrated anomaly evidence for heterogeneous operational trajectories under noisy, incomplete and changing conditions.	Two documented benchmark systems; anomaly taxonomy and dataset cards; preregistered evaluation protocol; FDA monitoring module; calibrated thresholds.	WP1–WP2 / D1–D2 / MS1–MS2.
O2: Graph-functional detection and localisation	Integrate functional evidence with physical, logical or communication dependencies using self- and semi-supervised graph models to detect and localise node-, edge- and subgraph-level anomalies.	Graph-functional detector; at least eight baselines; topology-sensitivity, leave-subsystem-out and ablation analyses.	WP3 / D3 / MS3
O3: Trustworthy Operational Risk Intelligence	Quantify uncertainty and explanation reliability and integrate validated anomaly evidence into propagation-aware, human-supervised investigation priorities.	ORI prototype; calibration, fidelity and stability evidence; propagation analysis.	WP4 / D4 / MS4
O4: Robustness and transferability	Determine the robustness and transferability of GRAIL-ORI across two complementary cyber-physical water-system benchmarks and identify the conditions under which its detection, localisation and ORI outputs remain reliable.	Cross-system validation report; robustness and transferability evidence; expert assessment; toolkit; demonstrator; reproducibility and implementation guidance.	WP5 / D5 / MS5

The objectives are measurable and realistically achievable within 24 months because the architecture is modular and each analytical layer has a predefined fallback. Core validation will use BATADAL, which is openly available, together with one complementary research benchmark selected from SWaT, WADI or an equivalent cyber-physical water-system dataset. Access requests for the SWaT and WADI benchmarks through iTrust (SUTD) will be submitted in M1. The complementary benchmark will be selected according to access conditions, topology information, event annotations, sensor/actuator coverage and complementarity of operating conditions. If access to the preferred benchmark is not secured by MS1, an equivalent open or research-accessible dataset will be activated, ensuring that validation never depends on a single access decision. At least eight functional, statistical, one-class, reconstruction, temporal and graph-based baselines will be compared under temporally ordered evaluation. Primary evaluation criteria and success thresholds will be preregistered at MS1, before final model comparison. They will include event-level detection performance, false-alarm rate at matched sensitivity, localisation performance and calibration. GRAIL-ORI will be considered to provide meaningful added value when the integrated graph-functional approach achieves, on at least one benchmark, either (i) a $\geq 10\%$ relative reduction in false-alarm rate at matched sensitivity, or (ii) a $\geq 5\%$ relative improvement in event-level detection or localisation performance over the strongest reproducible non-relational baseline, without material deterioration in calibration. Secondary analyses will assess detection delay, robustness to missing data and topology perturbations, computational cost and explanation fidelity and stability. Statistical uncertainty and paired comparisons will be reported where appropriate. These thresholds provide an ex-ante definition of meaningful improvement while allowing negative or non-superior results to remain scientifically informative.

GRAIL-ORI goes beyond the state of the art not by introducing functional analysis, graph learning, uncertainty quantification or explainability in isolation, but by integrating them into a quantitatively validated anomaly-intelligence chain for networked cyber-physical systems. Continuous operational trajectories are represented functionally rather than reduced only to fixed-dimensional observations; infrastructure dependencies are explicitly used for detection, localisation and propagation analysis; uncertainty is carried through the analytical pipeline; and explanations are evaluated for fidelity and stability before contributing to operational prioritisation. The resulting ORI layer therefore provides not only an anomaly score, but a calibrated and traceable evidence structure describing

affected components, relational propagation and investigation priorities under human oversight. The ambition of GRAIL-ORI is to establish when this integrated approach provides reliable added value over strong non-relational and graph-based alternatives, and equally importantly, to identify the conditions under which it does not.

1.2 *Soundness of the proposed methodology (including interdisciplinary approaches, consideration of the gender dimension and other diversity aspects if relevant for the research project, and the quality of open science practices).*

Overall methodology. GRAIL-ORI uses a layered, benchmark-driven workflow linking operational trajectories to human-supervised anomaly intelligence. It assumes exploitable temporal structure, documented or inferable infrastructure relations and scarce/uncertain labels. Functional representations and calibrated monitoring evidence become graph attributes for detection and localisation; uncertainty and explanation reliability are assessed before validated evidence enters ORI. All data-dependent preprocessing, graph inference and model fitting use training periods only; each layer has a fallback.

Methodological layer	Methods, assumptions and role in the project	Objective
1. Data and governance	Benchmark governance and dataset cards; verification of licences and access conditions; separation of natural, controlled and synthetic events; temporally ordered splits; preregistered evaluation protocol.	O1
2. Functional monitoring	Splines/basis expansions, wavelets or functional principal components [1,2]; functional statistics; bootstrap/conformal calibration alternatives evaluated out of sample.	O1
3. Graph-functional detection	Functional evidence becomes graph attributes; documented physical, logical or communication relations are preferred, complemented by similarity or sparse learned graphs when needed. Self-/semi-supervised prediction, reconstruction and message passing are compared with non-relational alternatives [3–5]; learned graphs use training data only.	O2
4. Uncertainty and explanations	Bootstrap/conformal uncertainty and functional, graph, counterfactual or temporal attribution; calibration, fidelity and perturbation-stability tests [8]. Unreliable explanations are withheld from ORI.	O3
5. ORI	Calibrated anomaly, localisation, propagation, uncertainty and validated explanation evidence will be combined through a transparent, pre-specified evidence-aggregation rule to generate affected-component evidence and ranked investigation priorities under human oversight; sensitivity analysis will assess ranking robustness to alternative aggregation choices.	O3-O4

Validation and human assessment. Validation will use the complementary cyber-physical benchmarks defined in Section 1.1. Temporally ordered validation, leave-subsystem-out tests, ablations, missing-data and topology perturbations will assess leakage, robustness and transfer. Between M18 and M22, 8–12 operators/domain experts will compare conventional alerts with ORI outputs using localisation, decision time, usefulness, clarity, workload and calibrated trust. This provides structured usability evidence, not population-level generalisation. Participation requires prior ethics/data-protection approval and consent; responses will be pseudonymised. If recruitment is insufficient, at least six experts will complete qualitative cognitive walkthroughs.

Methodological challenges and contingencies. Limited access/labels are addressed through research-accessible alternatives and self-/semi-supervision, with dataset replacement at MS1. Uncertain topology is handled by comparing documented and inferred graphs and retaining relational evidence only when stable under ablations/perturbations. Overfitting is controlled through temporal validation, regularisation, at least eight baselines and ablations; at MS3, the strongest reproducible statistical, hybrid or graph model is retained. Unstable scores are recalibrated; unreliable explanations are simplified or withheld.

Expert availability may constrain the human assessment of O3–O4. This challenge is addressed through recruitment across multiple UDC, MODES and CITIC channels, a target panel of 8–12 operators or domain experts, and a

predefined fallback of at least six experts completing structured cognitive walkthroughs, subject to prior ethics and data-protection approval.

Interdisciplinary integration. Functional statistics preserves temporal structure; graph learning models dependencies/propagation; statistical inference provides calibration/uncertainty; cybersecurity and infrastructure knowledge guide interpretation; software engineering/HPC supports reproducibility. Integration is operational: functional evidence becomes graph attributes, graph structure changes localisation, and uncertainty reaches the ORI layer.

Gender and diversity. Core data are machine-generated telemetry and system events, so sex/gender are not relevant variables in the anomaly models. Diversity matters in expert assessment and transfer: recruitment will seek variation in role, seniority, discipline and gender where practicable, without subgroup inference from the small sample. Infrastructure configuration, operating regime and sensor density will be considered where they affect transferability. Only minimum participant metadata will be collected.

Open science and research-output management. The evaluation protocol and primary criteria will be preregistered. Code, permitted splits, configurations, logs, model/dataset cards, containers and documentation will be version-controlled; publications and reusable non-sensitive outputs will be openly accessible with appropriate licences and persistent identifiers. A FAIR Data Management Plan will be delivered in M6 and updated in M24. Restricted data/topologies/vulnerabilities will follow “as open as possible, as closed as necessary”, using metadata, synthetic alternatives or controlled access where appropriate.

Storage, retention and preservation requirements will be assessed proportionately in the Data Management Plan according to the size and access conditions of the selected benchmark datasets and the volume of derived functional representations, model outputs, checkpoints, logs and reproducibility objects. Storage use will be monitored during implementation, and unnecessary intermediate outputs will be removed while preserving the research objects required for reproducibility. Existing UDC/CITIC computing and scalable-storage infrastructure will support these requirements.

1.3 Quality of the supervision, training and of the two-way transfer of knowledge between the researcher and the host

Supervision and scientific fit. Prof. Salvador Naya Fernández, Full Professor in Statistics and Operations Research at UDC and member of MODES/CITIC, will supervise GRAIL-ORI. His expertise in FDA, statistical process monitoring, nonparametric inference, reliability, industrial statistics and sensor-based monitoring directly supports the project’s statistical foundations. He previously supervised the researcher’s PhD and has competitive and industrial project experience, including EDINAF with NAVANTIA, providing a sound basis for scientific guidance and knowledge transfer. GRAIL-ORI extends beyond their earlier joint work towards graph learning, trustworthy AI, cybersecurity validation and ORI, supported by complementary expertise within CITIC.

Supervision. Fortnightly meetings will review science, training, ethics, outputs and risks; monthly MODES/CITIC discussions will provide complementary technical feedback. Stage gates occur at M4, M8, M12, M18 and M22, with Career Development Plan reviews at M6, M12, M18 and M24. Prof. Naya ensures scientific coherence and career guidance, while the researcher leads graph-functional and ORI design, implementation and documentation, supporting progressive independence.

Training domain	Competences and implementation	Evidence and contribution
Graph-functional AI and research software	Dynamic graphs, anomaly detection, self-supervision, uncertainty/explainability, cyber-physical monitoring, testing, containers and HPC workflows, M2–M20, supported by CITIC technical discussions,	Validated components, reproducible toolkit and independent research line.

	specialised seminars, software sprints and hands-on code-review sessions.	
Research and innovation management	MSCA management, integrity, ethics/data protection, open science, IP/transfer, leadership and grant writing, M1–M20.	European-project management and follow-on proposal capacity.
Communication, mentoring and career	Scientific writing, stakeholder/public engagement, mentoring and proposal development, M3–M24.	Presentations, mentoring and five-year career/funding strategy.

Two-way knowledge transfer. UDC/MODES/CITIC will transfer graph modelling, trustworthy AI, cybersecurity validation, scalable computing and EU project/IP practice through supervision, seminars, software sprints and code reviews. The researcher will transfer FDA, functional monitoring, bootstrap/nonparametric inference, anomaly detection and reproducible R/Python through an internal tutorial by M8, reusable monitoring components, benchmark protocols and seminars. This exchange connects his established statistical expertise with the competences required for O2–O4. No formal secondment or non-academic placement is needed; all objectives and deliverables are achievable at UDC.

Advanced graph-learning competences will be developed through the complementary AI, data-science and intelligent-network expertise available within the CITIC environment, using technical discussions, specialised seminars, software sprints and code reviews integrated into the project. This institutional training pathway complements Prof. Naya’s statistical supervision while avoiding dependence on a single specialist and can be adapted to the relevant expertise and training opportunities available during the fellowship.

1.4 Quality and appropriateness of the researcher’s professional experience, competences and skills

Miguel Alfonso Flores Sánchez is Associate Professor in Mathematics at Escuela Politécnica Nacional and holds a PhD in Statistics and Operations Research from UDC. His established expertise in FDA, statistical process monitoring, bootstrap/nonparametric inference and anomaly detection directly supports O1, while subsequent work in spatio-temporal anomaly detection, graph learning and IoT cybersecurity provides a strong basis for O2–O3. Relevant outputs include first-author contributions to functional control-chart methodology [2] and recent graph-neural-network research, as well as open statistical software and infrastructure-related anomaly and risk applications. His experience in reproducible R/Python development, interdisciplinary research, postgraduate supervision and project leadership supports implementation and knowledge transfer. GRAIL-ORI complements this profile with advanced graph-functional detection, trustworthy AI, scalable computing and cybersecurity validation, providing the competences required to establish an independent research line in graph-functional anomaly intelligence.

2. Impact

2.1 Credibility of the measures to enhance the career perspectives and employability of the researcher and contribution to his/her skills development

GRAIL-ORI supports a targeted transition from established expertise in functional monitoring, statistical inference and applied data science towards an independent research profile in trustworthy graph-functional intelligence for cyber-physical systems. The main competence gaps are advanced dynamic graph-functional learning for cyber-physical anomaly detection, uncertainty/explanation assessment, scalable research software, cybersecurity validation and European research/innovation leadership. These will be addressed through the training, supervision and project responsibilities described in Section 1.3, with the researcher leading ORI design and the open toolkit. An initial career-needs analysis in M1 will inform the Career Development Plan delivered in M6 and reviewed in M12, M18 and M24. Progress will be evidenced through three manuscripts, at least two conference contributions, software leadership, interdisciplinary/stakeholder engagement and a five-year research/funding agenda. By M24, the researcher will submit or bring to submission-ready stage at least one competitive follow-on proposal, with an ERC

Consolidator Grant or a Ramón y Cajal application considered as priority routes subject to the applicable eligibility conditions and call calendar. The resulting profile strengthens employability for academic leadership and R&D roles in statistics, AI, cybersecurity and infrastructure monitoring.

2.2 Suitability and quality of the measures to maximise expected outcomes and impacts, as set out in the dissemination and exploitation plan, including communication activities

An integrated dissemination, exploitation and communication strategy will operate from M1 and continue beyond the fellowship through publications, repositories, licences, networks and follow-on activities. Early actions will establish audiences, messages, channels and stakeholder engagement; the formal Dissemination and Exploitation Plan in M9, updated in M23, will consolidate responsibilities, Key Exploitable Results (KERs), IP/security decisions, exploitation routes and indicators.

Purpose and audiences	Measures and messages	Indicators
Scientific dissemination: FDA, statistical monitoring, graph learning, anomaly-detection and trustworthy-AI communities.	Three open-access manuscripts, ≥ 2 conference contributions, preprints, seminars and reproducible benchmark/software materials. Core message: continuous and relational evidence should be jointly modelled and quantitatively calibrated.	3 manuscripts; ≥ 2 conference contributions; 2 seminars; released reproducibility objects; documented scientific access, reuse or feedback.
Exploitation and transfer: infrastructure operators, technology centres, research-intensive firms, public organisations and UDC/CITIC transfer services.	Assess four KERs for maturity, user need, IP/security and route to use; conduct ≥ 3 stakeholder consultations; release demonstrator/technical guidance; practitioner workshop M20–M22. Routes may include collaborative pilots, services, licensing, training or follow-on projects according to KER maturity.	4 KER assessments; ≥ 3 consultations; ≥ 20 workshop participants; ≥ 1 documented stakeholder-validated exploitation or follow-on route by M24.
Communication: students, citizens and wider AI/data and European–Latin-American networks.	Project webpage by M2, hosted through an appropriate UDC/CITIC institutional channel and maintained by the researcher under WP6; accessible articles/visual explainers; public demonstration and dissemination through UDC/CITIC, EPN and professional channels. Core message: trustworthy infrastructure AI requires uncertainty awareness, reproducibility and human oversight.	≥ 4 accessible outputs; ≥ 40 participants across public activities; ≥ 1 Europe–Latin America activity, including ≥ 15 Latin-American participants.

The four KERs are: (1) graph-functional representation and benchmark protocol; (2) modular detection/localisation library; (3) uncertainty, robustness and explanation-evaluation toolkit; and (4) human-supervised ORI demonstrator and implementation guidance. For each KER, the M9/M23 plans will record ownership, maturity, target users, access conditions, protection and route to use. Background remains with its owners. Non-sensitive outputs will default to open-source/open-access release with appropriate licences; confidential know-how, controlled licensing or patent protection will be considered only when justified before disclosure. Security-sensitive topologies, vulnerabilities or exploit-enabling outputs will undergo appropriate UDC assessment and may use controlled access, redaction, delayed release or responsible disclosure.

2.3. The magnitude and importance of the project's contribution to the expected scientific, societal and economic impacts

Scientific impact. GRAIL-ORI will provide researchers in functional statistics, anomaly detection, graph learning and cyber-physical security with a reproducible framework for jointly analysing continuous telemetry and infrastructure dependencies while assessing calibration, localisation, robustness and explanation reliability. Its main scientific contribution is to establish whether graph-functional integration provides reproducible added value over strong statistical, temporal and graph-based alternatives and to identify the conditions under which this advantage holds or fails. Reusable benchmark protocols, evaluation procedures and software will facilitate independent comparison, extension and replication beyond the fellowship.

Technological and economic impact. Infrastructure-monitoring teams, technology centres and research-intensive organisations may use the project’s modular methods and demonstrator to assess anomaly localisation, uncertainty, propagation and investigation priorities before operational deployment. Reusable software, documented requirements and validated limitations can reduce the technical effort required to evaluate or adapt these methods for new monitoring settings. The project will not claim deployment-scale cost savings without operational evidence; instead, its direct contribution is a validated proof-of-concept and a documented route towards stakeholder assessment, collaborative pilots, technical services or follow-on innovation.

Societal impact. In the validated cyber-physical water-system context, better calibrated and more interpretable anomaly evidence can support earlier and more proportionate investigation of faults, communication disturbances and potentially malicious events affecting service continuity. Human operators remain responsible for interpretation and response. The direct beneficiaries are monitoring and infrastructure specialists; the longer-term societal value lies in supporting more reliable evidence for protecting essential services. Transfer to energy, transport, industrial or digital infrastructures is a potential longer-term pathway rather than a direct project impact and will depend on subsequent validation and uptake.

The impact pathway is explicit: calibrated and explainable anomaly evidence, reusable software and the demonstrator constitute the project outputs; improved localisation and investigation prioritisation are the expected operational outcomes; over the longer term, these can support more resilient continuity of essential water services as societal impact.

3. Quality and Efficiency of the Implementation

3.1 *Quality and effectiveness of the work plan, assessment of risks and appropriateness of the effort assigned to work packages*

The 24-month plan comprises six overlapping and modular Work Packages: WP1 establishes data and evaluation; WP2 functional monitoring; WP3 graph-functional detection; WP4 Trustworthy ORI; WP5 cross-system validation and release; and WP6 coordinates training, management and impact. Stage gates at M4, M8, M12, M18 and M22 support timely fallback decisions. The 24 person-months are allocated as WP1: 3 PM; WP2: 5 PM; WP3: 5 PM; WP4: 5 PM; WP5: 4 PM; WP6: 2 PM. Training, documentation and impact activities are embedded in the scientific WPs and coordinated through WP6.

The 24-month duration is justified by the sequential dependence of the main methodological stages: benchmark and evaluation readiness must precede calibrated functional monitoring; this evidence then feeds graph-functional detection and, subsequently, trustworthy ORI integration. The overlapping schedule allows each layer to be validated before integration while reserving M15–M23 for cross-system testing, expert assessment, robustness analysis and reproducible release. A shorter action would substantially compress independent validation and transfer assessment, whereas the proposed duration preserves methodological depth without adding non-essential scope.

WP	Timing	Effort	Main tasks	Deliverables and milestones
WP1: Data, governance and benchmark foundation (O1)	M1-M4	3 PM	Dataset access/governance; anomaly taxonomy; dataset cards; graph definitions; temporally ordered splits; preregistered evaluation protocol.	D1: Benchmark, anomaly taxonomy and evaluation protocol, M4 . MS1: Data and evaluation readiness confirmed, M4 .
WP2: Functional representation and calibrated monitoring (O1)	M2-M9	5 PM	Functional/multiresolution preprocessing; functional monitoring; calibration and uncertainty; comparison with statistical and temporal baselines.	D2: Functional-monitoring module and manuscript draft, M9 . MS2: Functional layer meets predefined calibration and reproducibility criteria, M8 .

WP3: Graph-functional anomaly detection (O2)	M5-M14	5 PM	Graph construction; self-/semi-supervised modelling; node-, edge- and subgraph-level detection/localisation; comparison with graph and non-relational baselines; topology sensitivity and ablations.	D3: Graph-functional detector and benchmark report, M14 . MS3: Predefined model-selection criteria met and architecture selected for integration, M12 .
WP4: Trustworthy ORI (O3)	M10-M20	5 PM	Calibration; component, relational, variable and temporal explanations; explanation-reliability testing; propagation evidence; ranked investigation priorities; expert-assessment protocol.	D4: Calibrated and explainable ORI framework and manuscript draft, M20 . MS4: Prototype meets predefined reliability criteria for expert assessment, M18 .
WP5: Cross-system validation, demonstrator and open release (O4; supports O3)	M15-M23	4 PM	Cross-system validation and validation manuscript; missing-data/topology stress tests; expert assessment; toolkit and demonstrator; transfer, security and reuse documentation.	D5: Toolkit, demonstrator and cross-system validation report, M23 . MS5: Validation and release evidence completed, M22 .
WP6: Training, management and impact (cross-cutting O1–O4)	M1-M24	2 PM	Project coordination; supervision/CDP; ethics and data management; D&E&C coordination; reporting and follow-on planning.	D6.1: Mobility Declaration, M1 . D6.2: Data Management Plan, M6 , update M24 . D6.3: Career Development Plan, M6 , reviews M12 , M18 and M24 . D6.4: Dissemination and Exploitation Plan, M9 , update M23 . D6.5: Evaluation Questionnaire, M24 .

Deliverables are tangible research, software or management outputs; milestones are verifiable decision points activating predefined fallbacks when required. Effort may shift from optional model variants or secondary analyses towards validation and reproducibility without changing the total 24 PM or compromising O1–O4.

Work package	M1-2	M3-4	M5-6	M7-8	M9-10	M11-12	M13-14	M15-16	M17-18	M19-20	M21-22	M23-24
WP1 Data/governance (O1)	•	• D1/MS1 M4										
WP2 Functional monitoring (O1)	● Start M2	•	•	• MS2 M8	● D2 M9							
WP3 Graph-functional detection (O2)			•	•	•	• MS3 M12	• D3 M14					
WP4 Trustworthy ORI (O3)					● Start M10	•	•	•	• MS4 M18	• D4 M20		
WP5 Cross-system validation (O4)								•	•	•	• MS5 M22	● D5 M23
WP6 Training & management (O1–O4)	• D6.1 M1	•	• D6.2/D6.3 M6	•	• D6.4 M9	• D6.3 review M12	•	•	• D6.3 review M18	•	•	• D6.4 M23; D6.2/D6.3/D6.5

Gantt chart. M indicates project month; • indicates activity during the complete two-month period; ● indicates activity during one month; D indicates deliverable; MS indicates milestone. No formal secondment or non-academic placement is included.

Risk assessment and mitigation. Risks are reviewed monthly and at stage gates M4, M8, M12, M18 and M22. Corrective action is triggered by unresolved benchmark access, >1-month expected deliverable delay, failure of predefined performance/reliability criteria, delayed ethics approval or unsustainable computational demand. Likelihood/severity values below represent residual risk after the methodological design measures described in Section 1.2 and the preventive measures implemented throughout the project.

Critical risk	Likelihood / severity	Mitigation and contingency
R1: Delayed data access, rare events or uncertain labels	M / H	Prioritise open and research-accessible benchmarks; submit access requests in M1; use modular data adapters and self-supervised or semi-supervised learning; distinguish natural, controlled and synthetic events. Replace an inaccessible dataset with an open equivalent at MS1.
R2: Incomplete, uncertain or changing topology	M / M	Compare physical, logical, communication-based, similarity-based and sparse learned graphs; quantify topology sensitivity; retain relational evidence only when stable under predefined perturbations. Use a simplified graph or the strongest non-relational model otherwise.
R3: Graph-functional models do not outperform or transfer	M / H	Apply temporal validation, leave-subsystem-out testing, regularisation, at least eight baselines and ablation studies. At MS3, select the strongest reproducible statistical, hybrid or graph model, even when the most complex graph architecture is not retained.
R4: Calibration or explanations fail reliability criteria	M / M	Compare bootstrap and conformal alternatives; evaluate calibration, fidelity and perturbation stability. Recalibrate unstable scores and simplify or withhold explanations that do not meet the predefined criteria.
R5: Ethics approval or expert recruitment is delayed	L / M	Prepare ethics and data-protection documentation before M12; submit it sufficiently early; recruit through multiple UDC, MODES and CITIC channels; collect no participant data before approval. If recruitment is insufficient, use the six-expert qualitative cognitive-walkthrough fallback.
R6: Scope, integration, computation or security constraints threaten completion	L / H	Use modular interfaces, sparse computation, profiling, controlled access and UDC/CITIC computing resources. Freeze optional architectures or secondary cases and prioritise the core benchmark, validated detector, ORI framework, toolkit and demonstrator. Sensitive outputs will undergo security and responsible-disclosure assessment before release.

3.2 *Quality and capacity of the host institutions and participating organisations, including hosting arrangements*

Hosting arrangements and integration. Universidade da Coruña (UDC) will be the sole beneficiary and host of GRAIL-ORI. From M1, the researcher will be integrated into MODES and the CITIC research environment under Prof. Salvador Naya Fernández. Fortnightly supervision, monthly technical discussions, group meetings, seminars, training and software/reproducibility reviews will provide continuous feedback while supporting progressive independence and leadership of the graph-functional and ORI components.

Scientific and technical capacity. MODES provides expertise in FDA, statistical process monitoring, nonparametric inference, reliability and sensor-based monitoring, directly supporting O1 and statistical validation. CITIC complements this with AI/data science, intelligent networks, cybersecurity and high-performance computing

capabilities required for O2–O4. Available UDC/CITIC resources include CPU/GPU computation, scalable storage, software-development and version-control environments, digital libraries and institutional repositories, sufficient for benchmarking, graph-model training, robustness analyses and development of the toolkit/demonstrator. UDC institutional services will support researcher onboarding and mobility arrangements, project management, ethics and data protection, open science and data management, and IP and exploitation activities throughout the fellowship.

Host-researcher complementarity. The researcher contributes established expertise in FDA, statistical monitoring, anomaly detection, nonparametric inference and reproducible R/Python software; UDC/MODES/CITIC provides the complementary graph learning, trustworthy AI, cybersecurity, HPC and European research/transfer environment required for O2–O4. This combination enables immediate contribution to O1 while supporting acquisition of the competences needed for scientific independence. No associated partner, formal secondment or non-academic placement is included; all objectives, deliverables and milestones can be completed at UDC.

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